Kalinga Institute of Industrial Technology
School of Computer Engineering

SAP ERP — Project Report

Order-to-Cash (O2C): Complete Sales Cycle

(SAP SD Module — ECC 6.0)

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1. Executive Summary

This project report documents the end-to-end Order-to-Cash (O2C) business cycle implemented in SAP ERP (ECC 6.0) for a fictitious textile manufacturer, NovaTex Industries Pvt. Ltd., headquartered in Lucknow, Uttar Pradesh.

The O2C cycle is one of the most commercially critical processes in any product-based enterprise. It represents the complete journey of a customer's order — from the moment a sales inquiry lands on the desk, all the way through delivery, billing, and finally the receipt of payment. Every rupee that flows into the business passes through this pipeline.

In this report, we have configured a realistic SAP SD environment for NovaTex Industries, mapped real-world business requirements to SAP customizing steps, and documented each transaction code with process narratives. The goal is to demonstrate not just how to click through SAP, but to understand why each step exists — the business logic behind the system.

Company Name	NovaTex Industries Pvt. Ltd.
Industry	Textile Manufacturing & Wholesale Distribution
SAP Module	SAP SD (Sales & Distribution) — ECC 6.0
Process Covered	Order-to-Cash (O2C) — Full End-to-End Cycle
Org. Structure	1 Company Code | 1 Sales Org | 2 Distribution Channels
Estimated Duration	12 to 15 working days to implement fully
Prepared For	KIIT SAP 6th Semester Project Submission

2. Company Overview — NovaTex Industries Pvt. Ltd.

Before we dive into transaction codes and configuration menus, it helps to know who we are building this system for. NovaTex Industries is our fictitious protagonist — a mid-sized Indian textile company that has decided to go live with SAP to replace its fragmented legacy systems.

2.1 Company Background

NovaTex Industries was founded in 2008 and has grown steadily into a wholesale supplier of cotton, synthetic blended fabrics and readymade garments. It sells to retail chains, boutique exporters, and institutional buyers such as schools and hospitals that procure uniforms in bulk.

Before SAP, NovaTex ran on a patchwork of Tally for accounts, Excel sheets for orders, and a homegrown warehouse tracking tool. Order mix-ups were frequent, and reconciling monthly books took a week. The decision to adopt SAP was driven largely by two pain points: stock-outs on fast-moving fabric SKUs, and delayed collections from distributors.

2.2 Organizational Structure in SAP

SAP requires a formal organizational hierarchy before any business transaction can be processed. Below is the structure we have defined for NovaTex in the system:

SAP Org Element	NovaTex Value	Business Meaning
Company Code	NT01	Legal entity for accounting
Sales Organization	NT10	Responsible for selling products
Distribution Channel	10 — Wholesale	Bulk B2B sales to distributors
Distribution Channel	20 — Direct Retail	Sales to end-consumer shops
Division	01 — Fabrics	Product line categorization
Sales Area	NT10 / 10 / 01	Primary sales area used in this project
Plant	NT01 — Lucknow Warehouse	Storage & delivery point
Shipping Point	NTSP	Physical dispatch location

2.3 Master Data Setup

Two types of master data are absolutely foundational to the O2C process: the Customer Master and the Material Master. Without these, SAP cannot process a single sales document.

Customer Master — Transaction: XD01

NovaTex's primary customer for this project is Raipur Garments Wholesale Ltd. — a B2B buyer based in Chhattisgarh. The customer master is created in three views:

- General Data: Name, address, communication details, tax numbers (GST)
- Company Code Data: Reconciliation account, payment terms (Net 30), credit limit (INR 15,00,000)
- Sales Area Data: Sales district, pricing procedure, shipping conditions, order combination settings

Material Master — Transaction: MM01

The product sold in our scenario is Cotton Drill Fabric (50 metre rolls), Material Number: NT-FAB-001. The material master spans multiple organizational views, each maintained by a different department:

- Basic Data 1: Material description, base unit of measure (ROL = Roll), division
- Sales: General/Plant: Loading group, delivering plant, availability check rule
- Sales: Sales Org. 1: Delivering plant, tax classification for GST
- MRP 1: Reorder point, lot size — relevant for availability checks
- Accounting 1: Valuation class, standard price for financial posting

Condition Records / Pricing — Transaction: VK11

Pricing in SAP SD is handled through a Pricing Procedure. For NovaTex, we use a custom procedure ZNTTX with the following key condition types:

Condition Type	Description	Value	Applicability
PR00	Base Selling Price	INR 480 / metre	Customer / Material specific
K007	Customer Discount	3%	Wholesale channel only
JGST	GST (5% Fabric)	5% on net value	Tax — auto-determined
KF00	Freight Charge	INR 200 / roll	Only for orders > 100 km

3. The Order-to-Cash Cycle — What It Is and Why It Matters

The Order-to-Cash cycle is the series of business steps that begins when a customer decides they want to buy something from you and ends when the money lands in your bank account. Every step in between — the inquiry, the negotiation, the order confirmation, the warehouse pick, the truck departure, the invoice, the payment reminder — is part of this cycle.

In SAP SD, this cycle is represented as a chain of interconnected documents. Each document flows into the next, creating a paper trail that connects a business conversation ("I want 500 rolls") to an accounting entry ("Revenue: INR 2,40,000"). The power of SAP is precisely this link — nothing falls through the cracks because every step leaves a document, and every document references its predecessor.

3.1 The Seven-Step O2C Process Flow

Here is the complete O2C flow as it will be executed for NovaTex Industries:

Step 1	**Pre-Sales: Inquiry (IN) — T-Code: VA11** *Customer calls or emails asking about product availability and pricing. SAP captures this as an Inquiry document.*

Step 2	**Pre-Sales: Quotation (QT) — T-Code: VA21** *Based on the inquiry, NovaTex prepares a formal price quote. The quotation locks in the price for a validity period.*

Step 3	**Sales Order (OR) — T-Code: VA01** *Customer accepts the quote. A Sales Order is raised — this is the core document of the entire O2C process.*

Step 4	**Delivery Document — T-Code: VL01N** *Warehouse team picks, packs, and ships the goods. The Outbound Delivery records goods movement and inventory reduction.*

Step 5	**Goods Issue (GI) — T-Code: VL02N** *Stock is formally removed from inventory. This triggers the first FI (accounting) document — debit COGS, credit stock.*

Step 6	**Billing / Invoice — T-Code: VF01** *Customer invoice is generated. This posts revenue to the P&L and creates a receivable in the customer's account.*

Step 7	**Payment Receipt — T-Code: F-28** *Customer pays. The incoming payment is posted, clearing the open receivable and completing the cycle.*

4. Step-by-Step Process Documentation

This section walks through each step of the O2C cycle in detail — covering the business context, the SAP path, the key fields to be filled, and the document it produces.

4.1 Step 1: Creating the Sales Inquiry

Transaction Code: VA11 | Display: VA12 | Inquiry List: VA15

SAP Menu Path: Logistics → Sales & Distribution → Sales → Inquiry → Create

Raipur Garments calls NovaTex and asks: "We need 500 rolls of Cotton Drill for our summer collection. What's the price, and can you deliver by month-end?" This conversation is captured as an Inquiry in SAP.

Field	Value Entered	Why It Matters
Order Type	IN	Identifies this as an Inquiry document
Sales Organization	NT10	Determines which sales team handles this
Distribution Channel	10	Wholesale — impacts pricing & terms

Field	Value Entered	Why It Matters
Division	01	Product line — Fabrics
Sold-to Party	CU1001 (Raipur Garments)	Customer being served
Material	NT-FAB-001	Cotton Drill Fabric roll
Quantity	500 ROL	500 rolls as requested
Requested Del. Date	End of current month	Customer's expected delivery window

Output: SAP generates Inquiry Number IN-2024-0047. This number is logged and handed to the sales executive who will prepare the quotation.

4.2 Step 2: Creating the Sales Quotation

Transaction Code: VA21 | Change: VA22 | Display: VA23

Reference: Can be created with reference to Inquiry IN-2024-0047

The sales team now prepares a formal quotation for Raipur Garments. Creating the quotation with reference to the inquiry copies over all item data, avoiding re-entry and ensuring traceability.

- Validity Start: 1st of current month
- Validity End: 30 days from start date — after this the price is no longer binding
- Net Price per Roll: INR 480 (base) minus 3% wholesale discount = INR 465.60
- GST (5%): INR 23.28 per roll
- Freight: INR 200 per roll (Raipur is ~950 km — freight condition KF00 triggers)
- Total per Roll: INR 688.88 approx.
- Total Order Value (500 rolls): INR 3,44,440

The quotation is printed (or emailed via SAP output) to Raipur Garments. SAP document number: QT-2024-0031.

4.3 Step 3: Creating the Sales Order — The Core Document

Transaction Code: VA01 | Change: VA02 | Display: VA03

Document Type: OR (Standard Order)

Raipur Garments confirms the quotation. The sales team creates a Sales Order in SAP, referencing QT-2024-0031. This is the most important step — the Sales Order is the backbone document from which all subsequent steps flow.

Header Level — Key Fields

- Sold-to Party: CU1001 — Raipur Garments Wholesale Ltd.

- Ship-to Party: CU1001-SH — Raipur Garments, Warehouse Address (may differ from sold-to)
- Bill-to Party: CU1001-BL — Same entity for this customer
- Purchase Order No.: RGW/2024/PO/448 — Customer's internal PO number (mandatory for B2B)
- Requested Delivery Date: As agreed — SAP checks availability automatically

Item Level — Key Fields

- Material: NT-FAB-001 | Quantity: 500 ROL
- Plant: NT01 (Lucknow Warehouse) — determined automatically from customer-material info record
- Shipping Point: NTSP — determined from plant + shipping conditions
- Item Category: TAN (Standard Item) — triggers normal delivery and billing

Credit Check & Availability Check

SAP runs two automatic checks at the moment the Sales Order is saved:

- Credit Check: Current outstanding for Raipur Garments is INR 4,20,000. Credit limit is INR 15,00,000. The order value of INR 3,44,440 keeps them within limit — order proceeds without a credit block.
- Availability Check (ATP): SAP checks current stock (600 rolls in NT01 warehouse) minus already committed quantities (80 rolls for another order). Available: 520 rolls. Order of 500 can be confirmed. Confirmed delivery date is accepted.

Output: Sales Order OR-2024-6782 is created and confirmed.

4.4 Step 4: Creating the Outbound Delivery

Transaction Code: VL01N | Collective: VL10A | Change: VL02N

SAP Menu Path: Logistics → Sales & Distribution → Shipping & Transportation → Outbound Delivery → Create → Single Document

On the confirmed delivery date, the warehouse team receives the picking order. SAP creates an Outbound Delivery document with reference to the Sales Order.

Activity	T-Code / Function	What Happens
Create Delivery	VL01N	Delivery doc created, stock reserved for this shipment
Transfer Order (WM)	LT0A / LT1A	Warehouse mgmt. assigns storage bin for picking
Picking Confirmation	LT12	Picker confirms physical pick from Bin A-07-12

Activity	T-Code / Function	What Happens
Packing	VL02N → Pack	500 rolls packed into 25 pallets (20 rolls/pallet)
Shipment Creation	VT01N	Truck booking and route assignment (optional)

Output: Delivery Document 80016742 is created and picking is confirmed.

4.5 Step 5: Post Goods Issue (PGI)

Transaction Code: VL02N → Post Goods Issue button | Or: VL09 to reverse

Posting Goods Issue is the physical-to-legal moment: it is when the goods legally leave NovaTex's possession. This single click sets off a chain of automatic postings across the entire system:

- Inventory is reduced by 500 rolls in Plant NT01, Storage Location SL01
- Accounting document is created: Debit Cost of Goods Sold (COGS) | Credit Inventory Account
- Delivery status changes to 'Goods Issue Posted' — billing is now possible
- Material document is created for traceability in MM
- Controlling document updates product cost in CO-PA

Accounting Entry Generated: Dr. COGS Account (P&L) | Cr. Finished Goods Inventory — INR 2,10,000 (500 rolls @ INR 420 std. cost)

4.6 Step 6: Creating the Customer Invoice (Billing Document)

Transaction Code: VF01 | Collective Billing: VF04 | Display: VF03

Document Type: F2 (Standard Invoice)

With goods issued, the finance team raises the customer invoice. The billing document is created with reference to the delivery document and automatically picks up all item details, quantities, and pricing.

Invoice Component	Calculation	Amount (INR)
Base Price (PR00)	500 rolls × INR 480	2,40,000
Discount (K007)	3% on 2,40,000	- 7,200
Net Value	2,40,000 − 7,200	2,32,800
Freight (KF00)	500 rolls × INR 200	1,00,000
Taxable Value	Net + Freight	3,32,800

Invoice Component	Calculation	Amount (INR)
GST — CGST (2.5%)	2.5% of 3,32,800	8,320
GST — SGST (2.5%)	2.5% of 3,32,800	8,320
Total Invoice Value	Taxable + GST	3,49,440

When the billing document is saved, SAP automatically releases it to accounting (FI):

- Debit Accounts Receivable (Raipur Garments): INR 3,49,440
- Credit Revenue Account (Sales — Fabric): INR 2,32,800
- Credit Freight Revenue: INR 1,00,000
- Credit GST Payable (Output Tax): INR 16,640

Output: Billing Document 90035812 is created. Invoice is printed and dispatched to customer.

4.7 Step 7: Incoming Payment Posting

Transaction Code: F-28 (Incoming Payment) | Clearing: F-32

SAP Menu Path: Accounting → Financial Accounting → Accounts Receivable → Document Entry → Incoming Payment

30 days later (per Net 30 terms), Raipur Garments transfers INR 3,49,440 to NovaTex's bank account. The accounts receivable team posts this in SAP:

- Document Date & Posting Date: Date of bank credit confirmation
- Account: House Bank — HDFC Current Account (Bank Key: NTHD01)
- Amount: INR 3,49,440
- Open Item: Select Billing Document 90035812 for clearing

SAP automatically creates the accounting entry:

- Debit Bank Account (Asset): INR 3,49,440
- Credit Accounts Receivable (Raipur Garments): INR 3,49,440

The open item on Raipur Garments' customer account is now cleared. The O2C cycle for this transaction is complete. Aged receivables report (transaction FD10N / FBL5N) will now show this line item as fully settled.

5. Document Flow & Traceability

One of SAP's most powerful features is the Document Flow — a chain of linked documents that lets you trace any transaction from start to finish. In SAP, you can open any document (say, the invoice) and click the Document Flow button to see every preceding and succeeding document in the chain.

For NovaTex's transaction with Raipur Garments, the complete document flow looks like this:

Step	Document Type	Document Number	T-Code	Status
1	Inquiry	IN-2024-0047	VA11	Completed
2	Quotation	QT-2024-0031	VA21	Completed
3	Sales Order	OR-2024-6782	VA01	Completed
4	Outbound Delivery	80016742	VL01N	Completed
5	Material Document (GI)	4900021847	VL02N/PGI	Posted
6	Billing Document	90035812	VF01	Posted
7	FI Document (Invoice)	1400033210	VF01 (auto)	Cleared
8	Payment Document	1500044992	F-28	Cleared

To view this in SAP: Open any document (e.g., Sales Order via VA03) → Menu → Environment → Document Flow. The system displays the entire chain visually with green/yellow/red status indicators.

6. Key Customizing / Configuration Steps

The O2C cycle doesn't work out of the box — it needs to be configured to match the company's business rules. All configuration is done in SPRO (IMG — Implementation Guide). Below are the critical customizing steps for the NovaTex O2C setup:

6.1 Defining the Enterprise Structure

1. Create Sales Organization NT10: SPRO → Enterprise Structure → Definition → SD → Define Sales Organization
2. Create Distribution Channels 10 and 20: SPRO → ... → Define Distribution Channel
3. Create Division 01: SPRO → ... → Define Division
4. Assign Sales Org to Company Code NT01
5. Assign Distribution Channel to Sales Organization
6. Assign Division to Sales Organization
7. Set up Sales Area (NT10/10/01) — the combination of all three

6.2 Sales Document Type Configuration

SPRO → Sales & Distribution → Sales → Sales Documents → Sales Document Header → Define Sales Document Types

- Verify/configure document type IN (Inquiry): Number range, default delivery type, completion rule
- Verify/configure document type QT (Quotation): Validity period, reference requirements
- Verify/configure document type OR (Standard Order): Billing type F2, delivery type LF, item category determination

6.3 Pricing Procedure Configuration

8. Define Condition Tables: Which key fields determine price (Sales Org / Customer / Material)
9. Define Access Sequences: The search order for finding applicable condition records
10. Define Condition Types: PR00, K007, JGST, KF00 — each with its own settings
11. Define Pricing Procedure ZNTTX: Add all condition types in sequence with subtotals, requirements, and alternative base values
12. Assign Pricing Procedure: Tied to Sales Area + Customer Pricing Procedure + Document Pricing Procedure

6.4 Shipping & Delivery Configuration

- Define Shipping Point NTSP: SPRO → Logistics Execution → Shipping → Basic Shipping Functions → Shipping Points
- Assign Shipping Point to Plant: Plant NT01 → Shipping Point NTSP
- Define Shipping Point Determination Rules: Based on loading group + shipping condition + plant
- Configure Delivery Document Type LF: Number range, picking relevance, goods movement type 601

6.5 Billing Configuration

- Define Billing Document Type F2: SPRO → SD → Billing → Billing Documents → Define Billing Types
- Set up Billing Plan if needed (not used for standard O2C)
- Define Copying Controls: Delivery → Billing (VTFL) — specifies which fields copy from delivery to billing
- Account Determination: Revenue Account Assignment — links billing condition types to G/L accounts in FI

7. Reporting & Analysis in SAP SD

Once transactions are posted, SAP provides a rich set of standard reports to track sales performance, monitor open items, and analyse the business. Here are the key reports used in the NovaTex O2C context:

T-Code	Report Name	Business Use Case
VA05	List of Sales Orders	See all open/closed orders; filter by customer, date range
VF05	List of Billing Documents	Monitor all invoices raised in a period
VL06O	Outbound Delivery Monitor	Track pending deliveries and overdue shipments
FBL5N	Customer Line Items	View all open receivables and cleared items for a customer
FD10N	Customer Balance	Total balance and credit exposure per customer
VF45	Billing Due List	Deliveries ready to be invoiced — batch billing trigger
ME2M	Orders by Material	Cross-functional: see purchase orders for a material
MB52	Warehouse Stock List	Current stock levels at plant / storage location level

8. Implementation Challenges & Key Learnings

No SAP implementation — even a practice one — goes completely smooth. Here are some realistic challenges encountered (or anticipated) during the NovaTex O2C setup, along with the lessons drawn:

Challenge 1: Pricing Not Determined Automatically

Symptom: Sales Order shows zero price for material NT-FAB-001.

Root Cause: Condition record in VK11 was created for the wrong combination — Sales Org / Material only, but the access sequence expected Sales Org / Customer / Material.

Fix: Recreated condition record with the correct key combination. Also verified that the material's tax classification was correctly set in the material master (Sales Org view) to allow GST determination.

Challenge 2: Delivery Could Not Be Created

Symptom: VL01N throws error 'No open quantity found for delivery item.'

Root Cause: The Sales Order was saved with a delivery block (credit check had auto-blocked it due to a misconfigured credit limit).

Fix: Removed the delivery block in VA02 after verifying credit exposure in FD32. Reconfigured the credit check tolerance percentage in OVA8 for this customer risk category.

Challenge 3: Billing Document Not Posting to FI

Symptom: Billing document saves, but no FI document is created. Status shows 'Blocked for Accounting.'

Root Cause: Account determination (Revenue Account Assignment) was not configured. The condition type PR00 had no G/L account assigned in VKOA.

Fix: Set up account determination in VKOA by assigning the correct revenue G/L account (Sales — Fabric: 400100) to the chart of accounts + sales org + account key ERF combination.

Key Learnings

- SAP's document chain is unforgiving — if one step is misconfigured, everything downstream fails. Always validate with a test run before going live.
- Master data is the foundation. 80% of O2C issues trace back to incomplete or incorrectly configured customer master, material master, or condition records.
- Understanding the 'why' matters more than memorizing T-codes. Once you understand that VL01N creates a delivery because the goods need to be formally dispatched, the menu path becomes intuitive.
- Cross-module integration is SAP's real power — a single Goods Issue posting touches MM, FI, CO, and WM simultaneously. This is what makes SAP enterprise-grade.

9. Project Timeline — Estimated Effort

The following is a realistic effort breakdown for implementing and documenting the complete O2C cycle for NovaTex Industries, as a university project submission:

#	Activity	Duration	Deliverable
1	Company & Org Structure Setup in SAP	1–2 Days	Org chart + IMG screenshots
2	Master Data: Customer & Material	1–2 Days	Master data creation record
3	Pricing Configuration & Condition Records	1–2 Days	Pricing procedure document
4	End-to-End O2C Transaction Execution	2–3 Days	Step-by-step with T-codes
5	Document Flow Verification & Testing	1 Day	Document flow screenshots
6	Reporting & Output Testing	1 Day	Sample report outputs
7	Project Report Writing & Formatting	2–3 Days	This document
	TOTAL ESTIMATED EFFORT	12–15 Working Days	Complete Submission Package

Note: This timeline assumes access to an SAP training system (SAP GUI with a pre-configured client or IDES system). If the system needs to be set up from scratch, add 3–4 days for baseline client configuration.

10. Conclusion

The Order-to-Cash cycle is not just a series of SAP screens — it is the digital representation of how a business earns its revenue. Every click, every document, every automatic posting in the chain we've documented here corresponds to a real-world business event: a commitment made, goods moved, a bill sent, a payment received.

Through the lens of NovaTex Industries, we have walked through how SAP SD brings discipline, traceability, and real-time financial visibility to what would otherwise be a scattered, error-prone manual process. From the first inquiry to the last rupee collected — every step is documented, every department is connected, and every transaction is auditable.

What makes this exercise valuable beyond the grade is this: the concepts here — pricing procedures, document types, account determination, availability checks — are the same concepts that SAP consultants work with on real projects for real companies. The fictitious company changes; the framework doesn't.

Whether NovaTex is selling cotton drill fabric or a Fortune 500 company is shipping industrial equipment, the O2C logic in SAP remains constant. That universality is what makes SAP SD one of the most sought-after skills in the enterprise software world.

11. References & Further Reading

- SAP Help Portal — Sales and Distribution (SD) Module Documentation: help.sap.com
- SAP ECC 6.0 Configuration Guide for SD — SAP Press (Murali Narayanan & Ashish Sood)
- "SAP SD: Practical Guide" — 3rd Edition, SAP PRESS
- KIIT SAP Project Work Guidance Document — Course Instructor
- SCN (SAP Community Network) — community.sap.com — discussion threads on pricing procedure and account determination
- YouTube: "SAP SD Full Cycle" by STechies — useful visual walkthrough for beginners

Appendix: Quick T-Code Reference

T-Code	Function	Used At Stage
VA11/12/15	Inquiry: Create/Change/List	Pre-Sales
VA21/22/23	Quotation: Create/Change/Display	Pre-Sales
VA01/02/03	Sales Order: Create/Change/Display	Sales Order
VL01N	Create Outbound Delivery	Shipping
VL02N	Change Delivery / Post GI	Shipping / Goods Issue
VL10A	Collective Delivery Creation	Shipping (batch)
VF01/04	Create Billing / Collective Billing	Billing
VF02/03	Change/Display Billing Document	Billing
F-28	Post Incoming Payment	Finance / Collections
FBL5N	Customer Line Items Report	Finance / AR
VK11/12/13	Condition Records: Create/Change/Display	Pricing Setup
XD01/02/03	Customer Master: Create/Change/Display	Master Data
MM01/02/03	Material Master: Create/Change/Display	Master Data
SPRO	IMG Configuration Workbench	All Config Steps
VA05	Sales Order List	Reporting
VF05	Billing Document List	Reporting
FD32	Customer Credit Master	Credit Management
MB52	Warehouse Stock Overview	Inventory Check

— End of Report —

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